PROJECT TITLE

GhostLogin

Automated SSH Exposoure and Attack



Network Research

Program Code: NX201

OPERATION ORDER

Manual security assessments against large network ranges are slow, error-prone, and inconsistent. Repetitive authentication testing and post-access validation consume time and introduce operational mistakes. This project focuses on building a fully automated framework to identify exposed SSH services, validate access controls under controlled conditions, and execute predefined verification tasks without manual interaction.

1. VISION

Cyber units operating in an automated way.



2. MISSION

Design and implement an automated SSH assessment tool that validates network exposure, tests authorized credential sets in a controlled environment, and performs non-interactive post-access verification actions to confirm successful access.

3. STRATEGY

Fully automate the assessment process from target input to final reporting.
User interaction is limited to defining the scope and providing input data.
All remote actions run automatically without interactive sessions.

4. OBJECTIVES

Accept a IP range, validate it, and identify hosts exposing SSH services. The tool tests authentication using either built-in or user-provided credentials and, upon successful access, automatically executes a predefined verification command without opening an interactive shell.

Project Structure

1. **User Input and Validation**
   1.1. Prompt the user for an IP range or subnet to scan.
   1.2. Validate that the provided range is correctly formatted.

2. **Scanning for SSH Services**
   2.1. Scan the validated IP range specifically for hosts with SSH running.
   2.2. Collect and list all IPs that have an active SSH service.

3. Credential Brute Forcing

   3.1. Allow the tool to use either a built-in list of SSH credentials or accept a user-provided credentials file.
   3.2. Attempt to brute force SSH logins on the discovered hosts using these credentials.

4. Post-Exploitation Automation

   4.1. On successful login, run a predefined command on the remote machine non-interactively (for example, create a hidden file as a proof of concept).
   4.2. Do not open an interactive shell; all actions should be automated commands.
   4.3. Handle SSH password passing automatically without interactive prompts.
   4.4. Bypass host key verification prompts to ensure full automation.

5. Output and Reporting

At the end of the scan and brute force process, the script should display which IPs were successfully accessed and where the command was executed.

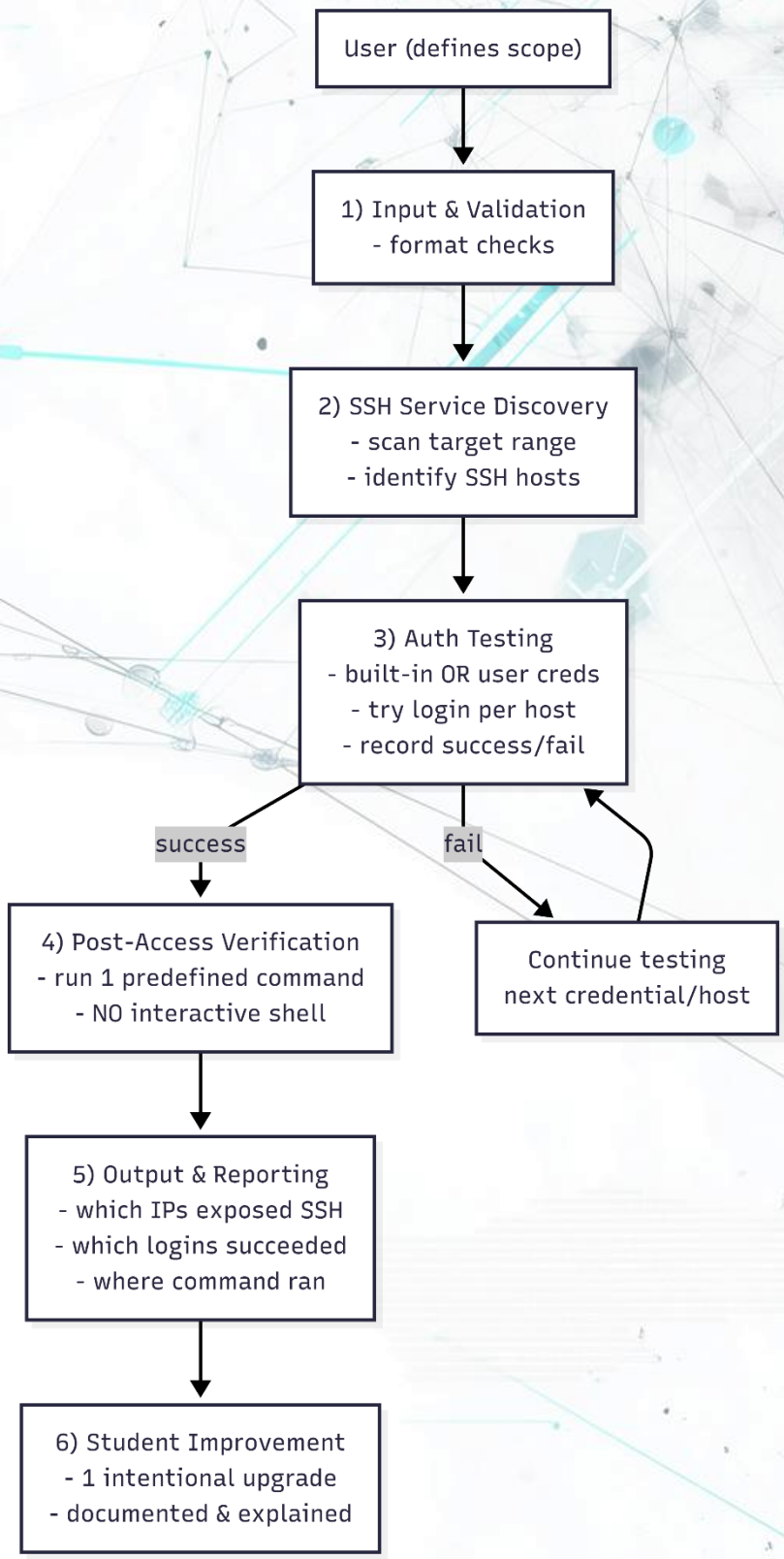
6. Creativity

Each student must add at **least one improvement** that they personally thought of. This can be related to design, functionality, automation flow, error handling, reporting, usability, or code structure. There is no required type of creativity. The only requirement is that the addition is intentional and explained. The project will not fail because of the creativity choice as long as one clear and original improvement is implemented and documented.

Comments

Use comments in your code to explain what you did.
If you are using code from the internet, add credit and links.
In the script, write the student's name and code, the class code, and the lecturer's name.

GhostLogin: Operational Logic Flow



Submitting Your Project

To ensure a successful submission of your project, please adhere to the instructions provided below. Following these guidelines will facilitate a smooth evaluation process and allow you to receive feedback on your work promptly.

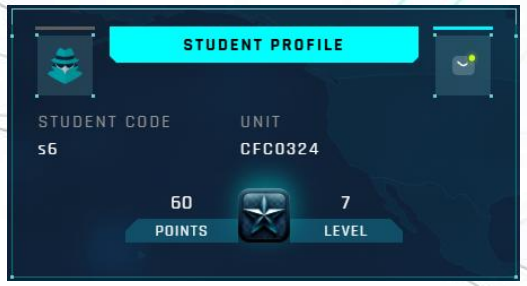
1. **What to Submit**

   Only .sh and .pdf files will be accepted. **No other file types** (e.g., .docx, .txt) will be accepted. Ensure your files are in the correct format before submission.

   You must submit both the script (.sh) and the corresponding PDF. The PDF should prove that you completed all the tasks and that the project works as expected. If you miss the script file or the PDF, you will be asked to resend your submission.

2. **File Naming**

   Ensure your project files follow the correct naming convention:
   UNIT.STUDENT.PROGRAM



   For example, if your unit is CFC0324, your student ID is s6, and your program code is nx201, the correct filenames should be:

   CFC0324.s6.nx201.sh
   CFC0324.s6.nx201.pdf

3. **Submission Process**

   Email Address: Forward your project materials to **projects@thinkcybergroup.com** and to your trainer.

   Email Subject: Format your email subject as follows: **Project: REMOTE CONTROL <Student Name:Unit Name>**. Make sure to replace **<Student Name>** with your actual name and **<Unit Name>** with the precise name of the unit as shown in the simulator.

4. **Unique Work**

   Your work **must** be unique. If the system detects that your project has been copied from another student, **both** the student who copied the work **and** the student who allowed their work to be copied will be **banned**. After being banned, the server will no longer check your future submissions.

Important: You are encouraged to help each other with ideas and techniques, but each student **must submit their own, original project**. If you submit a project that is flagged as identical or too similar to another student's, both of you will be banned from submitting future work to the system.

5. **Use of AI Tools**

   You are allowed to use tools like ChatGPT or other AI bots for assistance, but your program must work and be clear. If your AI code is simply copied, doesn't work, or is unclear, you will be banned from future submissions.

Response Time: After submitting, you can track the status of your submission (pass or fail) through the ***Progress Tracker*** inside your student account. Allow up to one week for your project to be reviewed.

How to Showcase Your Cybersecurity Projects on LinkedIn

1. **Write a Compelling Post**:
   When you share your project on LinkedIn, make sure to write a post that highlights the importance of the project and your key takeaways from it. Use engaging language and explain:
   - **The challenge** you solved.
   - **Tools** and technologies you used (mention relevant tools like Python, Shell Scripting, Scapy, etc.).
   - **Skills** you developed (problem-solving, incident response, penetration testing, etc.).
   - **Real-world relevance** of your project. Explain how it can be applied to secure networks, detect vulnerabilities, or analyze threats in a professional setting.

Example:

"Excited to share my latest project in Network Security (ZX305) where I built a script to detect network anomalies. This project helped me understand intrusion detection and how to safeguard systems against real-world threats. Using Python and Shell scripting, I developed a tool that can assist security teams in monitoring and reacting to suspicious network activities."

2. **Attach Your PDF**:
   Upload the PDF file of your project as an attachment to your post. This will allow recruiters, potential employers, or other professionals in the cybersecurity field to directly review your work.
   - Ensure your **PDF is professional**, free of errors, and clearly explains each section of the project.
   - **Highlight results**: If you were able to detect or simulate cyber threats, mention how your tool worked and the outcomes of your project.

3. **Record a Video Demo**:
   Videos are powerful on LinkedIn. Record a quick demo of your project where you:
   - Explain the problem.
   - Walk through your code.
   - Show your script in action (for example, showing how your network security tool detects a threat).

Example Intro:

"Here's a quick demo of my project where I developed a Python-based tool to monitor network traffic and detect potential cyber threats. Watch how it scans real-time traffic and flags suspicious activity."

4. **Mention the Certificate**:
   Once you complete the project, your simulator generates an official **ThinkCyber certificate**. Share this alongside your post to demonstrate your verified completion of the project.

Example:

"Proud to have completed my Network Security (ZX305) program with ThinkCyber! This certificate is proof of the skills I've acquired in identifying and mitigating cybersecurity threats."



Make sure to upload the certificate as an image. Use relevant hashtags like **#cybersecurity**, **#networksecurity**, and **#Python** to increase your visibility.

5. **Encourage Networking**:
   Mention that you are open to connecting with cybersecurity professionals and recruiters who might find your skills and projects relevant to their needs.